EXECUTIVE SUMMARY

PURPOSE OF THE APPARATUS

Snowed Under is a cooperative browser game in which up to eight players clear a shared, hexagonal Antarctic terrain. The architecture keeps one authoritative world while presenting each player with a responsive local view.

OPERATING PRINCIPLE

A deterministic simulation separates game rules from graphics and communications. In cooperative operation, a Node.js server advances that simulation at 30 steps per second. Browsers send input and commands, receive ordered updates, and render mirrors of the shared state with Three.js.

Local prediction makes movement and actions feel immediate; validation, correction epochs and replacement snapshots keep the mirrors aligned with the authority. In solo operation, the same core runs locally and saves progress in browser storage.

PRESERVATION AND CONTROL

Personal equipment is recovered by a persistent bearer credential, not by a display name. Saved records contain credential hashes. Atomic saves, validated compressed backups, build fingerprints and health responses support recovery and diagnosis.

BOUNDARIES AND NEXT DECISIONS

The present trust model is a friendly co-op group, not a moderated public room. Immediate operational priorities are disk capacity, external alerting, off-host backup automation and CI activation. Physical-device testing and human pacing trials remain necessary.

Reading guide: the specification follows, then five technical plates, illustrative claims, and further directions. This document describes release 0.2.0; proposed directions are not represented as implemented features.

UNITED STATES
PATENT APPLICATION

A technical illustration in the manner of the 1960s

APPARATUS FOR COOPERATIVE REMOVAL OF SIMULATED MATERIAL FROM A HEXAGONALLY DIVIDED TERRAIN

Snowed Under / Hexcavate

Architecture of release 0.2.0 • source e18de87

This specification describes a digital apparatus in which a plurality of operator stations participate in the clearing of one common terrain. A central calculating unit alone advances the cooperative simulation; each remote station maintains a corresponding display model.

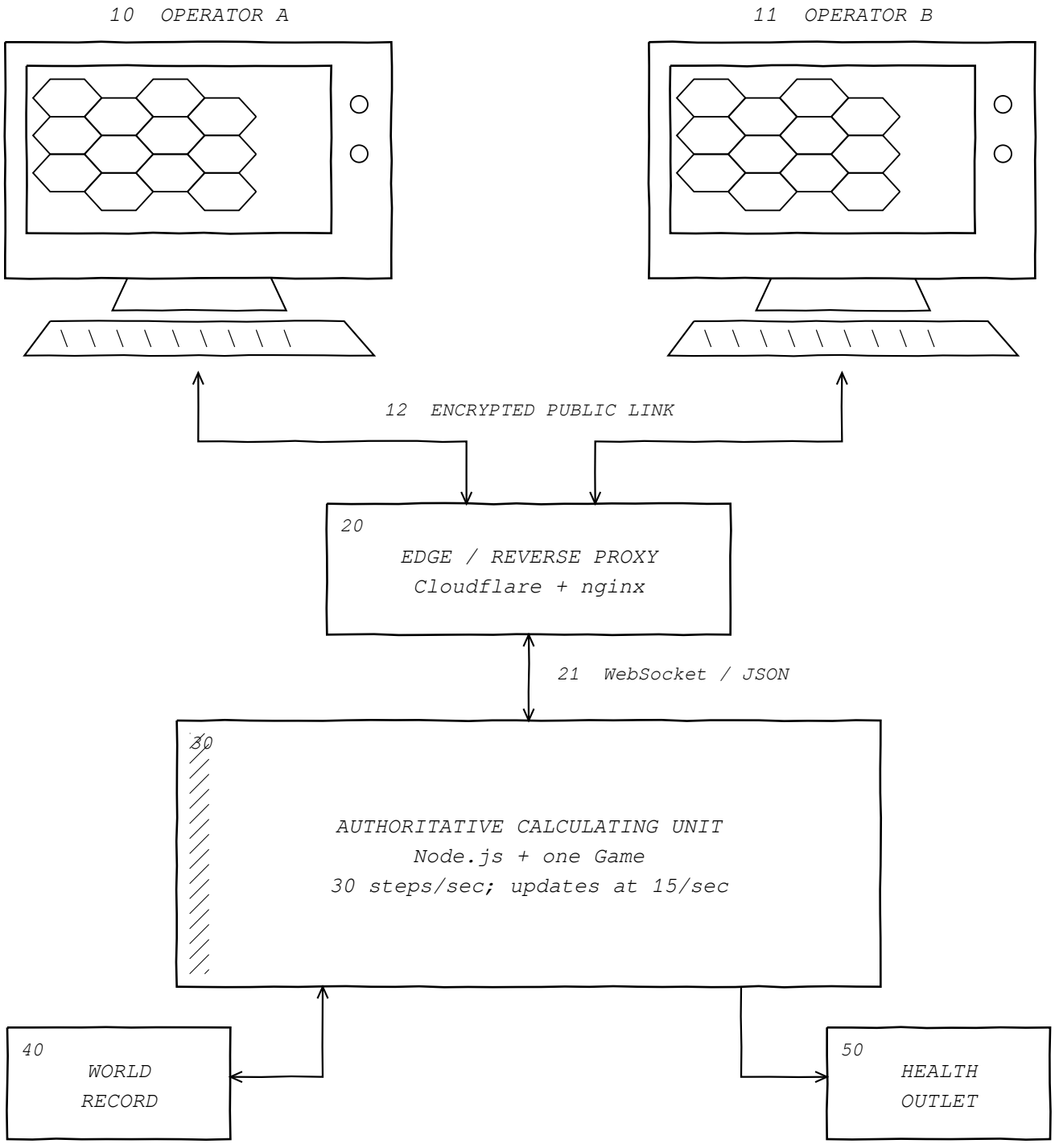
The preferred arrangement separates simulation, presentation and transport. Commands are admitted through a validating interface, whereas changes of terrain and movable bodies are returned as ordered records. A credential register preserves personal equipment without treating a display name as proof of ownership.

DRAWINGS

1. General arrangement and communication channels.
2. Terrain store and fixed-step calculating mechanism.
3. Operator station, display mirror and local prediction.
4. Ordered transmission and correction sequence.
5. Credential registry, durable record and supervision.

Editorial notice: a modern software architecture rendered in period patent-drafting style. This is neither a historical filing nor a representation of patentability or legal rights.

GENERAL ARRANGEMENT



10, 11: browser stations; up to eight connected players

FIG. 1

ONE SHARED WORLD; MANY CORRESPONDING VIEWS

TERRAIN STORE AND CALCULATING MECHANISM

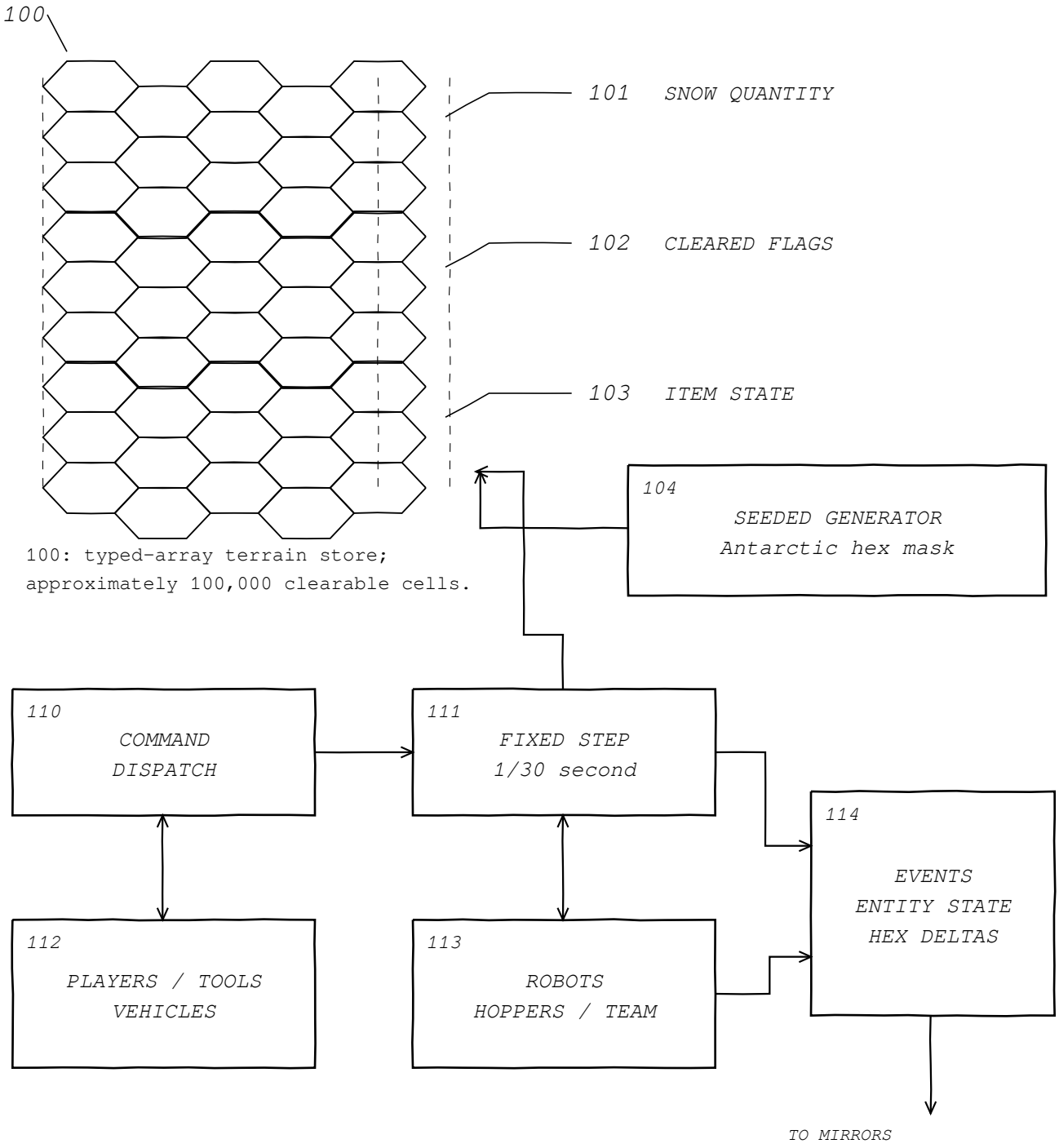
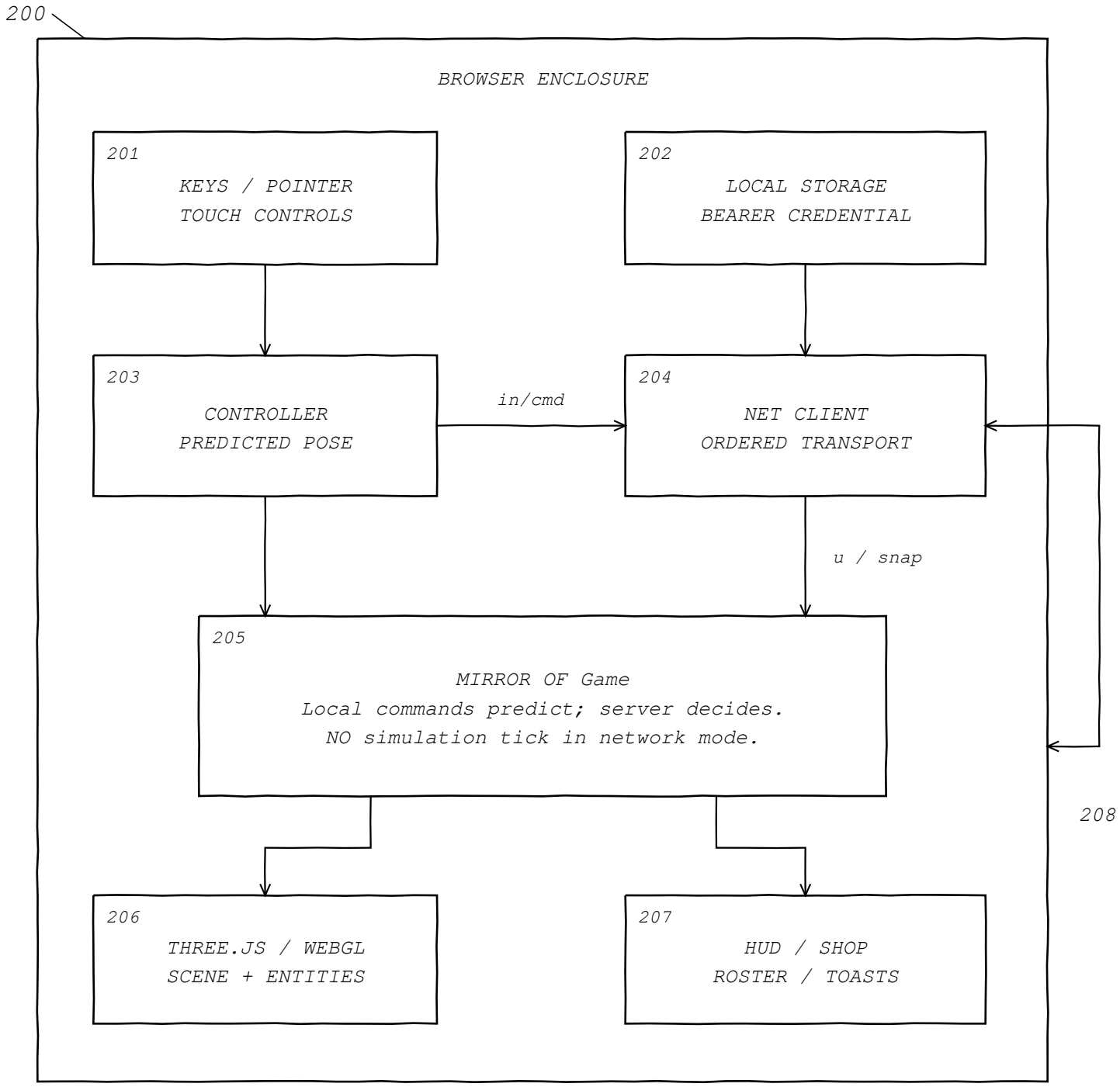


FIG. 2

STATE AND RULES ARE INDEPENDENT OF PRESENTATION

SECTION THROUGH AN OPERATOR STATION

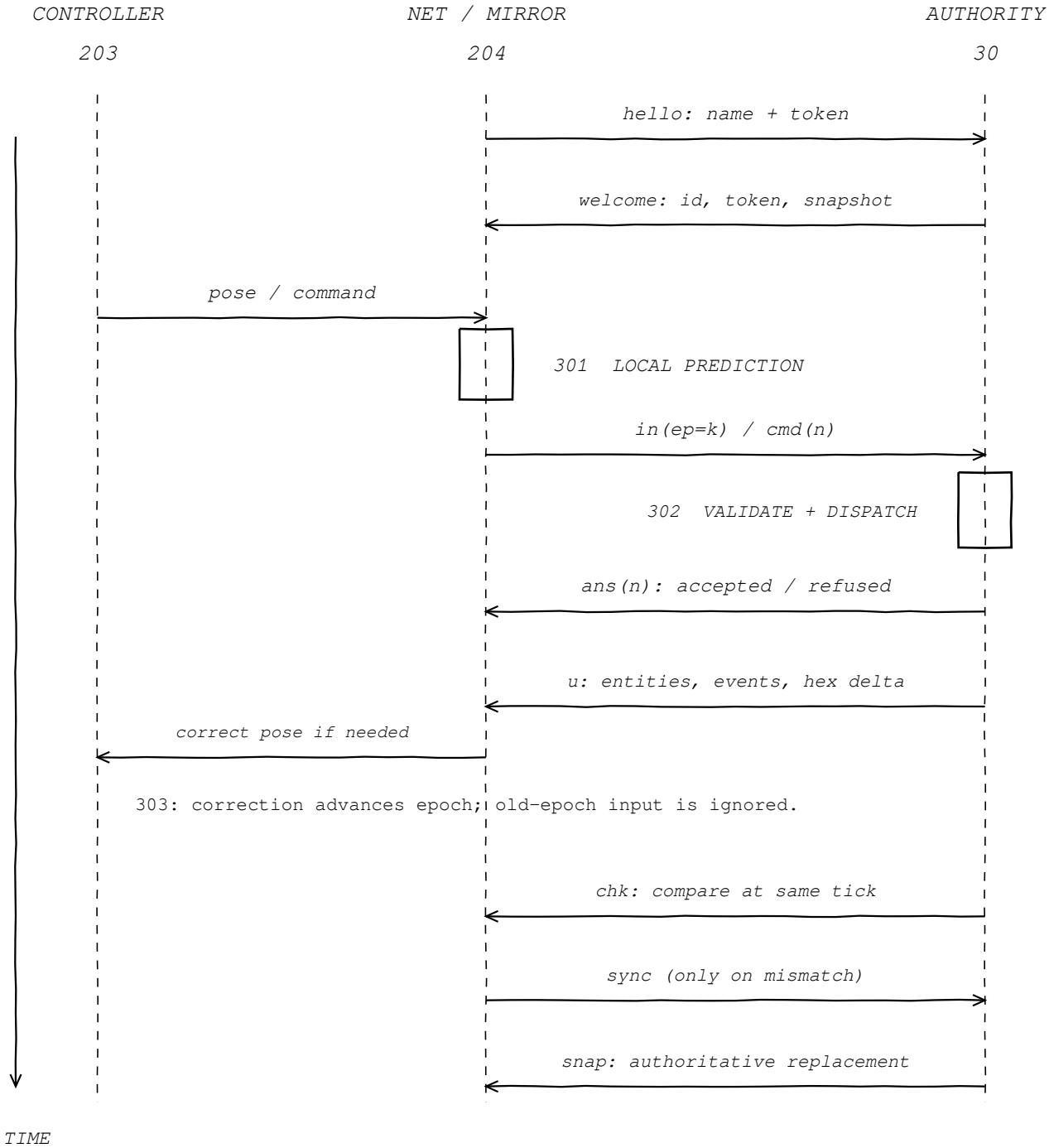


208: external WebSocket link to the authority (Fig. 1).

FIG. 3

PREDICT LOCALLY; RECONCILE TO THE CENTRAL RECORD

TRANSMISSION AND CORRECTION SEQUENCE



Schematic order; vertical distance is not elapsed time.

FIG. 4

SEQUENCED REPLIES, CORRECTION EPOCHS AND RESYNCHRONISM

IDENTITY AND PRESERVATION OF THE RECORD

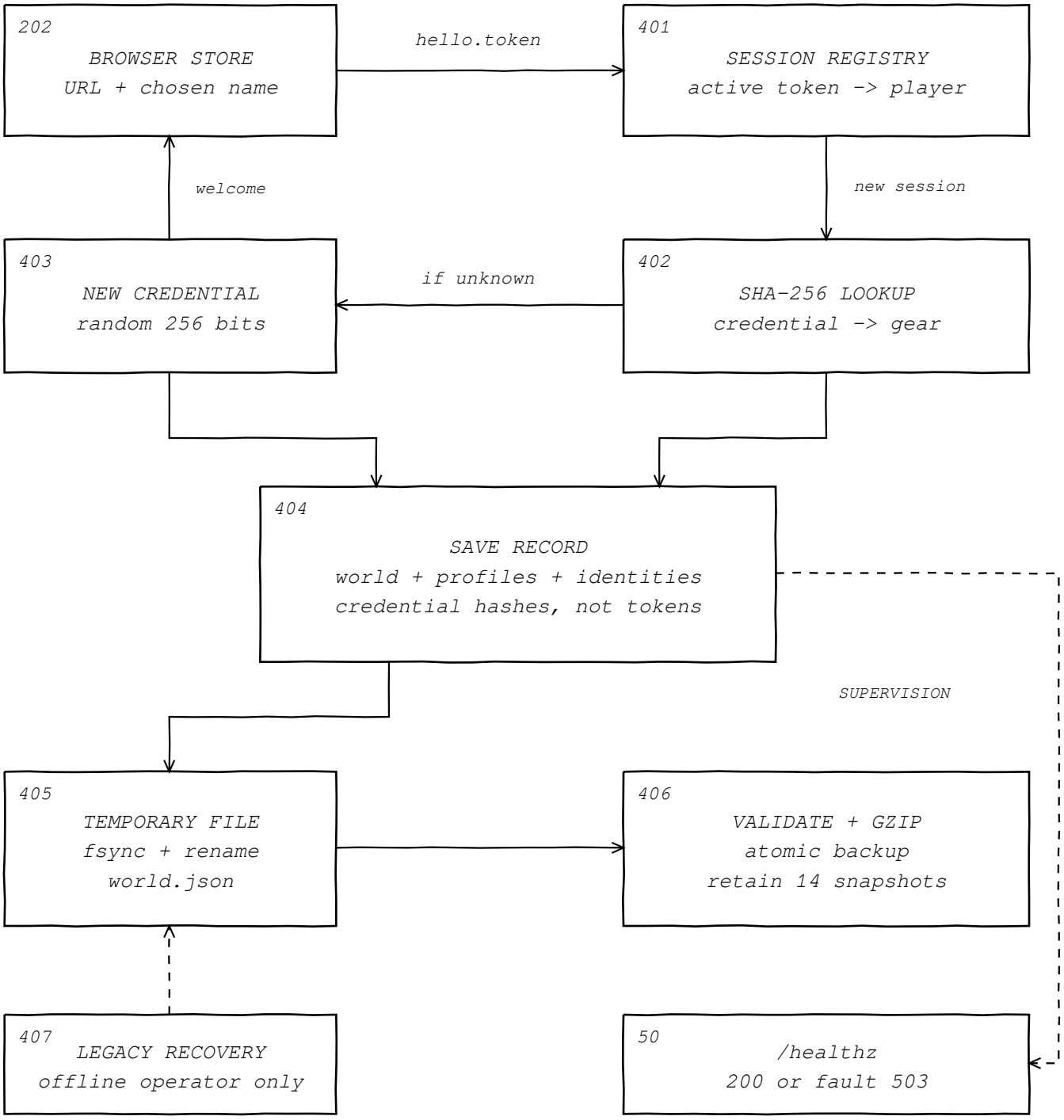


FIG. 5

PERSONAL OWNERSHIP IS NOT A DISPLAY NAME

DESCRIPTION AND ILLUSTRATIVE CLAIMS

Referring to Fig. 1, stations 10 and 11 communicate through proxy apparatus 20 with the central unit 30. Only unit 30 advances the shared simulation. The proxy serves the page and forwards the WebSocket channel; it does not decide game outcomes.

In Fig. 2 the terrain store 100 comprises snow quantities 101, cleared flags 102 and buried-item state 103. The seeded generator 104 determines the initial field. The fixed-step mechanism 111 operates the common rules and emits changed state through outlet 114.

Figs. 3 and 4 show that the mirror 205 is read by the same views used in solitary operation. Local pose and command predictions reduce apparent delay. Server answers, correction epochs and snapshots restore agreement. In solitary mode a local Game instead advances and is saved in browser storage.

Fig. 5 separates bearer credentials from saved credential hashes. A recognized hash restores personal gear after a restart; the visible name confers no ownership. Legacy profiles require offline operator recovery. Health outlet 50 reports server, simulation and save faults; the backup process is not itself an external alert service.

WHAT IS HEREIN ILLUSTRATED IS:

1. A cooperative simulation arrangement comprising a deterministic terrain model, one authoritative advancing mechanism, and a plurality of display mirrors receiving ordered state changes.

2. The arrangement of paragraph 1, wherein local prediction is reconciled by numbered answers, correction epochs, and replacement snapshots upon detected disagreement.

3. The arrangement of paragraph 1, further comprising a hashed credential registry and an atomically published save record with validated, bounded backup retention.

Source: src/core; src/client; src/net/protocol.js; server; deploy. Scope: trusted co-op, not public-room moderation. Claims above are illustrative prose only, not legal claims or a novelty opinion.

FURTHER DIRECTIONS

PROPOSED EXTENSIONS — NOT PRESENT CLAIMS

The following directions extend the apparatus described in the preceding sheets. They are recommendations for subsequent work, not a statement that these facilities are already installed or that any legal protection has been obtained.

I. OPERATIONAL RESILIENCE

Resolve the critically low VPS disk capacity without deleting unrelated data. Rotate the previously exposed administrative password through an agreed access procedure. Add external health and disk alerts with a defined human recipient.

II. RECOVERY AND RELEASE ASSURANCE

Automate encrypted off-host copies of validated backups, define a retention policy, and rehearse restoration on a separate instance. Authorize and activate the existing CI template; tie each future release to its tested source fingerprint.

III. ACCESS AND PERSONAL IDENTITY

Decide whether rooms remain trusted-group spaces or become public. For public use, specify invitations, shared-bank permissions and moderation before implementation. Design a deliberate cross-device credential transfer and recovery process.

IV. PHYSICAL-DEVICE AND HUMAN TRIALS

Exercise touch pickup, controls, reconnection and rendering on real phones and representative GPUs. Measure human progression and purchase choices across longer sessions before changing prices or making campaign-duration claims.

V. MEASURED SCALING

Profile mature worlds with multiple players, robots and hoppers. If traffic or rendering exceeds measured budgets, evaluate incremental robot-state patches and view updates while retaining deterministic core tests and mirror-agreement checks.

Acceptance: tested recovery, traceable releases, and measured user outcomes.